

Economics 8803: Economic Data Analysis

Problem Set 3

Due by: **March 21, 5 PM ET**

Instructions:

- Same as for Problem Set 1 and 2.
- Since this problem set includes simulation exercises, make sure to set the randomization seed such that your results are perfectly reproducible.

1 Empirical Problems

1. IV four ways: Consider the just-identified case with $J = K = 1$. Using any dataset of your choice and with any statistical software¹, verify that estimating IV using the following four different ways yields numerically equivalent results (do not worry about the violation of exclusion restriction).
 - (a) ‘Regular’ IV, i.e. $\tau_{IV} = (Z'X)^{-1}Z'Y$.
 - (b) Two Stage Least Squares: that is manually implement the two regressions (note that we don’t care about the standard errors for this problem), and find the 2SLS coefficient from the 2nd stage.
 - (c) The ‘Ratio’ IV, i.e. run the *reduced form* and the *first stage*, and take the ratio of the coefficients on Z .
 - (d) By solving the moment condition $E[Z_i, \epsilon_i] = 0$.
2. Many (Weak) IV: For an observed random sample of $n = 250$ observations $(y_i, x_i, z_i') \in \mathbb{R}^{1+1+l}$, consider:

$$\begin{aligned}y_i &= x_i\beta_0 + \epsilon_i \\x_i &= z_i'\pi_0 + u_i \quad (i = 1, \dots, n)\end{aligned}$$

1. Whatever dataset you use, please provide a simple summary table with key variables you’d use for the analysis (no need to explain/interpret it). Further, make sure to define the treatment variable, instrument, and outcome of interest. Again, no need to defend the validity of the instrument.

where, $l = 15, \beta_0 = 0, \pi_0 = (1, \dots, 1)' / \sqrt{n}$, and

$$(\epsilon_i, u_i)' \sim \mathcal{N} \left(0, \begin{bmatrix} 1 & 0.7 \\ 0.7 & 1 \end{bmatrix} \right)$$

$$z_i \sim \mathcal{N}[0, I_l], \quad (I \text{ is identity matrix})$$

(a) In large samples with multiple IVs the first-stage F -statistic is close to,

$$1 + \frac{n\mathbb{E}[(z_i' \pi_0)^2]}{l\sigma_u^2}$$

Calculate this value.

(b) In a statistical software of your choice (you may use the sample R code provided), generate data from the above model and calculate the following statistics: $\hat{\beta}_{OLS}, \hat{\beta}_{2SLS}, \hat{F}, \hat{\beta}_{JIVE}$.

Hint: $\hat{\beta}_{JIVE}$ is the Jackknife IV estimator (JIVE), which avoids overfitting of the first-stage by a leave-one-out procedure. Below are the steps to calculate $\hat{\beta}_{JIVE}$:

- For each i estimate the first-stage $\hat{\pi}_{-i}$ excluding observation i
- Obtain fitted values $\tilde{D}_i = z_i' \hat{\pi}_{-i}$ and set $\hat{\beta}_{JIVE} = (\tilde{D}' \tilde{D})^{-1} \tilde{D}' Y$ (note that in this question, x_i is the endogenous variable)
- Instead of running the first stage n times, you may calculate $\tilde{D}_i = z_i' \hat{\pi}_{-i} = \frac{z_i' \hat{\pi} - P_{ii} x_i}{1 - P_{ii}}$ where $P_{ii} = z_i' (\sum_{i=1}^n z_i z_i')^{-1} z_i$.
- Repeat (b) 10,000 times and report the simulated median of the coefficient estimates and the simulated mean of the F statistic. What do you observe?

(c) In the above simulation, verify that controlling the residuals from first stage as the 'control function' yields the same estimate as 2SLS (i.e. calculate the median control function estimate).

3. Measurement error: For a random sample of $n = 250$ observations, consider the following model:

$$y_i = D_i \beta_0 + \epsilon_i$$

where $\epsilon_i \sim \mathcal{N}(0, 1)$. You may choose whatever distribution for D_i .

- Consider, $\mathcal{D}_{1i} = D_i + \xi_i$, where $\xi_i \sim \mathcal{N}(0, \sigma_\xi^2)$.
- Consider, $\mathcal{D}_{2i} = D_i + \eta_i$, where $\eta_i \sim \mathcal{N}(0, \sigma_\eta^2)$.

You may choose whatever parameter values for β_0 , σ_ξ^2 , and σ_η^2 .

- (a) Regress Y_i on D_i and report the coefficient estimate.
- (b) Now, regress Y_i on D_{1i} and report the estimate. Show that this estimate is close to:

$$\beta_0 \frac{\text{Var}[D_i]}{\text{Var}[D_i] + \sigma_\xi^2}$$

- (c) Now, obtain the coefficient estimate from instrumenting $\mathcal{D}_{1i}$ with $\mathcal{D}_{2i}$.

Using a statistical software of your choice, repeat the above process 10,000 times and report the median of the coefficient estimates. What do you observe? Write a short note relating your findings to the model and theory presented in class.

Sample R Code

```
set.seed(123) # Ensure reproducibility
library("doParallel", quietly=TRUE)
nCores <- detectCores() #Run simulations parallel on multiple cores

nsim <- 10000
n <- 250
beta0 <- 0
cov <- .7
l <- 15
pi0 <- replicate(l, 1)/sqrt(n)

#Simulation
cl <- makeCluster(nCores); registerDoParallel(cl)
Result <- foreach(i=1:nsim,.combine=rbind) %dopar% {

  z <- t(replicate(n, rnorm(l))) # Instrument
  e <- rnorm(n) # errors
  u <- rnorm(n)
  x <- z %*% pi0 + sqrt(1-cov^2)*u + cov*e # Endogenous regressor
  y <- ...
  bOLS <- 0
  b2SLS <- 0
  bJIVE <- 0
  Fstat <- 0
  cf <- 0

  #Return statistics
  c(bOLS, b2SLS, bJIVE, Fstat, cf)
}
#cleanUp
stopCluster(cl); rm(cl)
#Report
c(median(Result[,1]),median(Result[,2]),median(Result[,3]),mean(Result[,4]), median(Result[, 5])
```